University of Murcia

Mathematics Department

Modelling Laboratory

Production Optimization at Los3Puentes

Grupo 1

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1 Introduction

This is the report for the group project in the Modeling Laboratory course, in which we detail how we identified the problem, what the problem consists of, the different considerations we took into account when formulating it, the formulation itself, the implementation in AMPL, and the conclusions from the entire process.

2 Problem

2.1 Finding the problem

At the beginning of the project, when our only task was to find a good problem to work on, we decided to split up so that each of us, following our professors' instructions, could ask our acquaintances and close contacts if they could help us.

We ended up finding several problems that seemed more or less promising, ranging from improving a high school schedule to reduce students' commuting, to attempts to optimize crop fields for different products in the fields of Yecla, as well as improving the distribution flow of a logistics company from the same town. Nevertheless, we decided to choose the problem proposed by the company Los3Puentes, whose CEO is the brother of the godmother of one of our group members. In addition, after visiting their facilities, we were very pleased with their willingness to help us and with the opportunity to solve a problem that would have a real impact in solving the company's difficulties.

2.2 Problem statement

Our problem consists in optimizing the production chain of the company "Los3Puentes" in order to maximize their profits. The production process runs as follows: each day the company sorts the artichokes by size, which is referred to as their caliber, and each type of caliber is then processed simultaneously and separately along the production line.

The production process is then divided into different stages in a specific order. It is important to note that not all products share the same stages. Finally, once this process is completed, the products are stored in warehouses and later sent to the different customers.

Our duty lies in avoiding the bottlenecks that are formed at the production stages shared by several products, since, as the head of the company explained to us, this was a problem that was limiting their production.

Focusing more on the specific situation of the factory, these are the characteristics we had to take into account to properly model the problem:

About the products:

- Each product goes through the same production chain but does not need necessarily to go through every stage.
- A product does not go through the same stage twice.

About the stages:

- Stages can be either operating or halted.
- Every operating stage has its own maximum production at which it can work.
- Some stages have a maintenance cost per hour regardless of whether they are operating, while others only have maintenance costs when they are operating.
- Among the stages which have a maintenance cost while operating, some depend on their rates while others do not.
- Some stages have an initial cost to be activated.
- No stage is permitted to process more than one product simultaneously.

About sales:

- The selling prices of each product will be predetermined for each customer. These prices are determined every week and can vary depending on multiple factors.
- Each customer is assigned a minimum demand that must be met and a maximum demand estimated by our company.

2.3 Considerations regarding the implementation

In this section, we detail the decisions we made to be able to implement the factory's operation in a model.

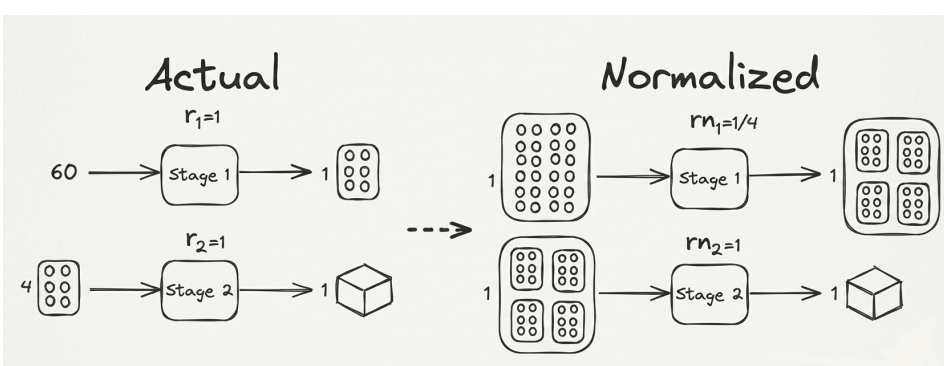
- Given a product, we call subproduct each item that is produced at a stage in the production chain. We refer to the rate of a stage as its production speed, that is, the number of subproduct units it produces per hour.

  This way of viewing the rates creates problems when checking whether a stage has enough subproduct from the previous stage to operate, since the rates depend on the subproduct.

  For this reason, we will consider that the rates are "normalized to the final product." That is, the rate of a stage tells us the quantity of final product that will be produced from the amount of subproduct generated at that stage. We make this adjustment by considering the unit of subproduct at each stage to be a batch containing the amount of subproduct required to produce one unit of the final product.

  Let us consider an example: our product in question is one box made up of 4 trays, each containing 6 artichokes. We divide the production into two stages: "Place in tray" and "Place in box," each with its corresponding rate (1 and 2, respectively).

  Since the second stage needs 4 trays to make 1 box, we take these 4 trays as the unit of subproduct. The normalized rate will be 1 when stage 2 produces 4 trays per hour, that is, when the real rate is 4.



In this way, we have real rates, which measure how many units of subproduct a stage produces per hour, and normalized rates, which measure how many blocks of subproduct a stage produces per hour. Our model will work with the normalized rates, calculated using the following formula:

$$rn_{p,e} = \frac{r_{p,e}}{C_{p,e}}$$

Where $rn_{p,e}$ is the normalized rate of stage e for product p, $r_{p,e}$ is the real rate and $C_{p,e}$ is the amount of subproduct that stage e needs for the production chain to produce one unit of product p.

- We will consider that the ratios can only be changed every hour. This is clearly a simplification over reality, given the fact that attempting to vary ratios continuously would involve non-linear restrictions. If a better approximation would be required, we would just need to consider as fractions of an hour our unit of time, without major changes to the model itself.
- We will consider that the stages do not need a minimum production to be activated; that is, a machine can be operating without producing anything.
- The entire production chain for a product must be completed within the same day, so by the end of the workday, no products can be left unfinished.
- For the implementation of the problem, we introduced a fictitious stage, the stage e_{fin}, which will be common to all products. The final stage e_{fin} does not represent any actual process. It just represents the finished product, stored and waiting to be sold.
- We assume that the factory operates the same amount of time every day.
- We assume that we can always meet the minimum demand. It is presumed that our company is prepared to satisfy its customers' demand and can produce in excess.
- Our model will assign the artichokes needed to meet the production's demand only at the beginning of the day.
- Maintenance costs do not depend on time, only on the type of product and the type of stage, as we specified earlier.
- In our implementation, the production chain can be stopped at any stage for any product, leaving the intermediate products stored in order to start processing artichokes for a different product; resuming the process for the first product whenever necessary.
- As a simplification, we will consider that the time it takes for the machines to be ready for production is negligible.
- The relationship between the cost of the stages that depend on the production rate and the rate itself is linear.

3 Formulation

We decided to first formulate a provisional, daily model and then extend it to a weekly form, since production occurs daily and each day the products are stored in the warehouses. In this way, at the end of the week, we will assign how much product we sold to each customer.

3.1 Daily formulation

In this provisional model, we will focus on the daily production stage, assuming that once the day is over, we will associate how much will be sold to each customer.

Sets and parameters: In this section we will describe all the sets and parameters that we will use in the model.

Let us define the sets by which our variables will be indexed.

- Our time units, hours, will be indexed by $HORAS := \{1, \ldots, h_{max}\}$.
- Products by $PRODS := \{1, \ldots, p_{max}\}$.
- Our clients will be $CLS := \{1, \ldots, c_{max}\}$.
- And the production stages are $ETAPAS := \{1, \ldots, e_{max}, e_{fin}\}$. The final stage e_{fin} will be represented by an index different from the rest. Thus, we can simply assign $e_{fin} := e_{max} + 1$. We will also use the following notation $ETAPAS^* := ETAPAS \setminus \{e_{fin}\}$.

Now we are in the position to define our parameters:

- The amount of artichokes we can make use of every day is: $CantAlc_g \geq 0 \quad \forall g \in DIAS$.
- The maximum production rate at which each stage can operate is given by the following matrix. $RatioMax := (R_{p,e})$ con $p \in PRODS, e \in ETAPAS^*$. $RatioMax_{p,e} \geq 0$.
- The maximum and minimum demand of each product assigned to each client is given by the following matrixes, $DemandaMin := (D_{p,c})$ with $p \in PRODS, c \in CLS$ and $DemandaMax := (M_{p,c})$ con $p \in PRODS, c \in CLS$ respectively. $M_{p,c} \geq D_{p,c} \geq 0$.
- The selling price of each product assigned to every client is given by the matrix: $PR := (PR_{p,c})$ with $p \in PRODS, c \in CLS$. $PR_{p,c} \geq 0$.

As we explained earlier, not all products go through every stage. We define then, for every product p, an ordered vector composed of the stages through which it passes. We will also need its dual concept, that is, the set of products which have to through a given stage e.

- $ORD_p := (e_{j_1}, \ldots, e_{j_{n_p}}, e_{fin})$ with $e_{j_i} \in ETAPAS$, $1 \leq i \leq n_p$ $\forall p \in PRODS$ ordered in increasing order. Let us also define $e_I(p)$ as the first element of ORD_p and $e_F(p) = e_{j_{n_p}}$ as its last element. Additionally, for convenience in the implementation of the constraints, we will write the following: $ORD_p^* := ORD_p \setminus \{e_{fin}\}$.
- $\Omega_e := \{p \in PRODS : e \in ORD_p^*\}$.
- For a specific p, given a $e_{j_i} \in ORD_p$ with $i \neq 1$, we refer to $e_{j_{i-1}}$ as $pred(e_{j_i})$.

We must also distinguish between the three types of stage, depending on their differences in their maintenance costs. This is a partition of the set $ETAPAS^*$. We can encounter "fixed maintenance" ($ETAPAS_f$), "binary maintenance" ($ETAPAS_b$) and "variable maintenance" ($ETAPAS_v$).

$$ETAPAS^* = ETAPAS_f \dot{\cup} ETAPAS_b \dot{\cup} ETAPAS_v$$

- The maintenance cost of the fixed stages does not depend on whether the stage is running or not. It also does not depend on the product or the production rate.
- Binary stages only incur a cost if the stage is active. The cost depends on the product but does not depend on the production rate.
- In the variable stages, maintenance cost is only dependent on the rate and product.

Another subset of the stages that we will have to define is "stages of initial cost" E_{ci} as all the stages with a cost when they are turned on.

Maintenance costs are given by the following matrixes:

- $Fijo = (F_e)$ with $e \in ETAPAS_f$, is the maintenance cost of the fixed stages.
- $Binario = (B_{p,e})$ con $p \in PRODS, e \in ETAPAS_b$, is the maintenance cost of the binary stages.
- $Variable = (V_{p,e})$ con $p \in PRODS, e \in ETAPAS_v$, is the maintenance cost of the variable stages when the rate is equal to 1.
- $CosteInicial = (CI_e)$ con $e \in ETAPAS_{ci}$ is the cost of turning on the stage e.

To sum up with our parameters, we define:

- The conversion factor from quantity of product to quantity of artichoke: $\Phi = (\Phi_p)$ with $p \in PRODS$, which is the amount of artichoke that is needed to produce p.

Variables: In this section, we specify and define the variables we will use along with their respective bounds.

- $c_{p,e,h}$ $\forall p \in PRODS, e \in ORD_p, h \in HORAS$: These represent the amount of product p waiting to be processed by the stage e at hour h. In this variable, we will use the auxiliary stage e_{fin} that we defined previously.

  The variable $c_{p,e_{fin},h}$ will be the product p already finished at hour h. Consequently, $c_{p,e_{fin},h_{max}}$ is the product finished in one day.

  It is important to note that $c_{p,e_I(p),1}$ will be the quantity of artichokes that we allocate to produce p, which we decide at the beginning of the day. In this way, $0 \leq c_{p,e_I(p),1} \leq \frac{CantAlc}{\Phi_p}$.

  The rest of variables $c_{p,e,h}$ are tied to that of the first hour, so the bound is not strictly needed; nevertheless, as our rates are normalized, we can set:

  $$0 \leq c_{p,e,h} \leq \frac{CantAlc}{\Phi_p} \quad \forall p \in PRODS, e \in ORD_p, h \in HORAS$$

- $r_{p,e,h}$ $\forall p \in PRODS, e \in ORD_p^*, h \in HORAS$: These are the amount of product p processed by the stage e at the hour h. They are bounded by the parameters previously defined: $0 \leq r_{p,e,h} \leq R_{p,e}$.
- $w_{p,e,h}$ $\forall p \in PRODS, e \in ORD_p^*, h \in HORAS$: These represent whether if the stage e is producing p or ot at time h. The variable is binary.

- $k_{e,h}$ $\forall e \in ETAPAS_{ci}, h \in HORAS$: It is binary and will be 1 if stage e has been turned on at hour h. Be careful not to confuse this variable with the previous one. The variable does not indicate whether the stage is active at hour h, but whither it has been activated at that hour, that is if it was off at $h-1$ and now it is on.

- $s_{p,c}$ $\forall p \in PRODS, c \in CLS$: These represent the quantity of product p sold to costumer c at the end of the day. In the next version they will represent weekly sales. $D_{p,c} \leq s_{p,c} \leq M_{p,c}$.

Objective function: Our objective is to maximize profits, which we will calculate as the revenue from selling the products minus the maintenance costs of the stages and the cost of starting certain stages.

$$Maximize: \quad Ingresos - Mantenimiento - CosteInicio$$

Where:

- $Ingresos$ is the amount of money we earn from selling products.

$$Ingresos = \sum_{p \in PRODS, c \in CLS} s_{p,c} PR_{p,c}$$

  This is, the sum over the sales of a product times its selling price.

- $Mantenimiento$ is the amount of money spent due to the maintenance of the stages.

$$Mantenimiento = \sum_{e \in ETAPAS_f} F_e \cdot h_{max} +$$

$$+ \sum_{p \in PRODS} \Bigg(\sum_{e \in ETAPAS_b \cap ORD_p^*} (\sum_{h \in HORAS} w_{p,e,h}) B_{p,e} +$$

$$\sum_{e \in ETAPAS_v \cap ORD_p^*} (\sum_{h \in HORAS} r_{p,e,h}) V_{p,e} \Bigg)$$

  This term is divided into three parts, one for every type of stage in which we divided our original set of stages. Let us note that the first term, the cost of fixed stages, is constant, so we could not include it in the objective function. But we decided to include it to improve readability. The second and third terms are those associated to the binary and variable stages. The costs of the binary stages depend on the variable $w_{p,e,h}$ and the costs of the variable stages on the rate of production.

- $CosteInicio$ is the amount of money spent by starting a stage that was previously stopped.

$$CosteInicio = \sum_{e \in ETAPAS_{ci}} (\sum_{h \in HORAS} k_{e,h}) CI_e$$

  We calculate it as the start-up cost for each time the stage has to be turned on.

Constraints: In this section, we will specify the constraints we have established to properly model our problem.

Production constraints: We describe the constraints that affect production.

• The quantity we send for production cannot exceed the raw materials available, which we model as:

$$\sum_{p \in PRODS} c_{p,e_I(p),1} \Phi_p \leq CantAlc$$

We need to convert the quantity of artichokes available into the quantity of product. Furthermore, we do this at the beginning of the day, as mentioned earlier.

• We must ensure that by the end of the day, nothing is left partially produced in the intermediate stages. Everything that has been introduced must be fully processed throughout the day. We model this with the following constraint:

$$c_{p,e_I(p),1} = c_{p,e_{fin},h_{max}} \quad \forall p \in PRODS$$

The constraint achieves its purpose since these variables represent the initial and final quantity of product p, respectively.

• There cannot be partially completed products at the beginning of the day. That is, the quantity of products in a stage that is midway through the process in the first hour must be 0:

$$c_{p,e,1} = 0 \quad \forall p \in PRODS, e \in ORD_p \setminus \{e_I(p)\}$$

This constraint could be omitted by not creating these variables, but the notation for the subsequent ones would become somewhat more obtuse. For better clarity, we have decided to keep it this way.

• Production is neither created nor destroyed; it is only transported. The stages cannot generate products out of nothing, nor can they lose products (assuming no errors in the production line). It is modeled as:

$$c_{p,e,h} = c_{p,e,h-1} - r_{p,e,h-1} + r_{p,pred(e),h-1} \quad \forall p \in PRODS, e \in ORD_p^* \setminus \{e_I(p)\}, h \in HORAS, h > 1$$

$$c_{p,e_{fin},h} = c_{p,e_{fin},h-1} + r_{p,e_F(p),h-1} \quad \forall p \in PRODS, h \in HORAS, h > 1$$

$$c_{p,e_I(p),h} = c_{p,e_I(p),h-1} - r_{p,e_I(p),h-1} \quad \forall p \in PRODS, h \in HORAS, h > 1$$

That is, for each hour, the quantity of product in a stage must be equal to the quantity from the previous hour minus what the stage produced during that hour, plus what was produced by the previous stage and delivered to it.

Although it is divided into three constraints, they all mean the same thing; it is just that the initial and final stages are special cases where one of the rates is not considered.

• No stage can produce over its storage:

$$r_{p,e,h} \leq c_{p,e,h} \quad \forall p \in PRODS \ \forall e \in ORD_p^* \ \forall h \in HORAS$$

• If a stage is stopped, its rate must be zero. This constraint models the desired operation for the variables $w_{p,e,h}$.

$$(w_{p,e,h} = 0) \implies (r_{p,e,h} = 0) \quad \forall p \in PRODS, e \in ORD_p^*, h \in HORAS$$

This logical constraint will be linearized later.

• Only one product can be processed in each stage at the same time:

$$\sum_{p \in \Omega_e} w_{p,e,h} \leq 1 \quad \forall e \in ETAPAS^*, h \in HORAS$$

As the variables w are binary, this constraint guaranties that only one of them can be on at once.

Fixed Cost Constraint: In this case, the two constraints are not linear; we will linearize them later. Both of them model when a machine goes from being off to being on.

• At the beginning of the day, at $h = 1$, the variable $k_{e,1}$ will simply be activated when some $w_{p,e,1}$ is also active. It is model as follows:

$$(\sum_{p \in \Omega_e} w_{p,e,1} = 1) \implies (k_{e,1} = 1) \quad \forall e \in ETAPAS_{ci}$$

• For any hour that is not the initial one, we need to check that some product is passing through the stage and that, in the previous hour, the stage was not occupied. This is:

$$(\sum_{p\in\Omega_e} w_{p,e,h} - \sum_{p\in\Omega_e} w_{p,e,h-1} = 1) \Longrightarrow (k_{e,h} = 1) \quad \forall h > 1, e \in ETAPAS_{ci}$$

We are only modeling one implication, since the variables $k_{e,h}$ affect negatively our objective function, so we can assume their default value to be 0.

Sales constraints:

• This constraint ensures that we cannot sell more or less than what we have produced.

$$\sum_{c\in CLS} s_{p,c} = c_{p,e_{fin},h_{max}} \quad \forall p \in PRODS$$

For each product, the sum of the quantities sold to each customer must be equal to the quantity produced.

Linearization: Some of the constraints that have arisen are not linear, so to make the problem easier to solve computationally, we will linearize them:

• For the constraint $(w_{p,e,h} = 0) \implies (r_{p,e,h} = 0)$ we set:

$$r_{p,e,h} \leq w_{p,e,h} N_{p,e} \quad \forall p \in PRODS, e \in ORD_p^*, h \in HORAS$$

Note that the rates $r_{p,e,h}$ are bounded by $R_{e,p}$, so it is convenient to take the big-M as: $N_{p,e} = R_{e,p}$.

• The other two constraints regarding the activation of the machines are linearized as follows:

$$\sum_{p\in\Omega_e} w_{p,e,1} \leq k_{e,1} \quad \forall e \in ETAPAS_{ci}$$

$$\sum_{p\in\Omega_e} w_{p,e,h} - \sum_{p\in\Omega_e} w_{p,e,h-1} \leq k_{e,h} \quad \forall h > 1, e \in ETAPAS_{ci}$$

Final formulation: By combining all the constraints, variables, and the objective function, our problem can be stated as:

$$
\begin{aligned}
\text{maximize} \quad & \sum_{p \in PRODS,\, c \in CLS} s_{p,c} PR_{p,c} - \sum_{e \in ETAPAS_f} F_e\, h_{max} \\
& - \sum_{p \in PRODS} \Bigg(\sum_{e \in ETAPAS_b \cap ORD_p} \Big(\sum_{h \in HORAS} w_{p,e,h} \Big) B_{p,e} \\
& + \sum_{e \in ETAPAS_v \cap ORD_p} \Big(\sum_{h \in HORAS} r_{p,e,h} \Big) V_{p,e} \Bigg) \\
& - \sum_{e \in ETAPAS_{ci}} \Big(\sum_{h \in HORAS} k_{e,h} \Big) CI_e
\end{aligned}
$$

$$
\text{subject to} \quad \begin{cases}
\bullet \sum_{p \in PRODS} c_{p,e_I(p),1} \Phi_p \leq CantAlc \\
\bullet c_{p,e_I(p),1} = c_{p,e_{fin},h_{max}} \\
\forall p \in PRODS \\
\bullet c_{p,e,1} = 0 \\
\forall p \in PRODS, e \in ORD_p \setminus \{e_I(p)\} \\
\bullet c_{p,e,h} = c_{p,e,h-1} - r_{p,e,h-1} + r_{p,pred(e),h-1} \\
\forall p \in P, e \in ORD_p^* \setminus \{e_I(p)\}, h \in HORAS, h > 1 \\
\bullet c_{p,e_{fin},h} = c_{p,e_{fin},h-1} + r_{p,e_F(p),h-1} \\
\forall p \in PRODS, h \in HORAS, h > 1 \\
\bullet c_{p,e_I(p),h} = c_{p,e_I(p),h-1} - r_{p,e_I(p),h-1} \\
\forall p \in PRODS, h \in HORAS, h > 1 \\
\bullet r_{p,e,h} \leq c_{p,e,h} \\
\forall p \in PRODS\ \forall e \in ORD_p^*\ \forall h \in HORAS \\
\bullet \sum_{p \in \Omega_e} w_{p,e,h} \leq 1 \\
\forall e \in ETAPAS^*, h \in HORAS \\
\bullet r_{p,e,h} \leq w_{p,e,h} R_{p,e} \\
\forall p \in PRODS, e \in ORD_p^*, h \in HORAS \\
\bullet \sum_{p \in \Omega_e} w_{p,e,1} \leq k_{e,1} \\
\forall e \in ETAPAS_{ci} \\
\bullet \sum_{p \in \Omega_e} w_{p,e,h} - \sum_{p \in \Omega_e} w_{p,e,h-1} \leq k_{e,h} \\
\forall h > 1, e \in ETAPAS_{ci} \\
\bullet \sum_{c \in CLS} s_{p,c} = c_{p,e_{fin},h_{max}} \\
\forall p \in PRODS \\
\\
\bullet 0 \leq c_{p,e,h} \leq \frac{CantAlc}{\Phi_p} \\
\forall p \in PRODS, e \in ORD_p, h \in HORAS \\
\bullet 0 \leq r_{p,e,h} \leq R_{p,e} \\
\forall p \in PRODS, e \in ORD_p^*, h \in HORAS \\
\bullet D_{p,c} \leq s_{p,c} \leq M_{p,c} \\
\forall p \in PRODS, c \in CLS \\
\bullet w_{p,e,h} \in \{0,1\} \\
\forall p \in PRODS, e \in ORD_p^*, h \in HORAS \\
\bullet k_{e,h} \in \{0,1\} \\
\forall e \in ETAPAS_{ci}, h \in HORAS \\
\bullet c_{p,e,h}, r_{p,e,h}, s_{p,c} \in \mathbb{Z}^+
\end{cases}
$$

3.2 Weekly formulation

In this section, we have added the set of days of the week, which adds an additional dimension to our variables. The fact that most constraints remain almost the same is due to the fact that most of them stem from the modeling of production, which is still daily. However, we find it useful to rewrite the formulation with the changes to clarify any small doubts and subtleties that may arise while reading the report. The most important change is that, in this case, sales are carried out at the end of the week.

Sets and parameters: In this section we will describe all the sets and parameters that we will use in the model.

Let us define the sets by which our variables will be indexed.

- Our time units, hours, will be indexed by $HORAS := \{1, \ldots, h_{max}\}$
- Products by $PRODS := \{1, \ldots, p_{max}\}$.
- Our clients will be $CLS := \{1, \ldots, c_{max}\}$.
- And the production stages are $ETAPAS := \{1, \ldots, e_{max}, e_{fin}\}$. The final stage e_{fin} will be represented by an index different from the rest. Thus, we can simply assign $e_{fin} := e_{max} + 1$. We will also use the following notation $ETAPAS^* := ETAPAS \setminus \{e_{fin}\}$.
- Our days of work are indexed by the set: $DIAS := \{1, \ldots, g_{max}\}$. We do not use the obvious choice of the letter d to represent the days as it already used for minimum demands.

Now we are in the position to define our parameters:

- The amount of artichokes we can make use of every day is: $CantAlc \geq 0$.
- The maximum production rate at which each stage can operate is given by the following matrix. $RatioMax := (R_{p,e})$ con $p \in PRODS, e \in ETAPAS^*$. $RatioMax_{p,e} \geq 0$.
- The maximum and minimum demand of each product assigned to each client is given by the following matrixes, $DemandaMin := (D_{p,c})$ with $p \in PRODS, c \in CLS$ and $DemandaMax := (M_{p,c})$ con $p \in PRODS, c \in CLS$ respectively. $M_{p,c} \geq D_{p,c} \geq 0$.
- The selling price of each product assigned to every client is given by the matrix: $PR := (PR_{p,c})$ with $p \in PRODS, c \in CLS$. $PR_{p,c} \geq 0$.

As we explained earlier, not all products go through every stage. We define then, for every product p, an ordered vector composed of the stages through which it passes. We will also need its dual concept, that is, the set of products which have to through a given stage e.

- $ORD_p := (e_{j_1}, \ldots, e_{j_{n_p}}, e_{fin})$ with $e_{j_i} \in ETAPAS$, $1 \leq i \leq n_p$ $\forall p \in PRODS$ ordered in increasing order. Let us also define $e_I(p)$ as the first element of ORD_p and $e_F(p) = e_{j_{n_p}}$ as its last element. Additionally, for convenience in the implementation of the constraints, we will write the following: $ORD_p^* := ORD_p \setminus \{e_{fin}\}$.
- $\Omega_e := \{p \in PRODS : e \in ORD_p^*\}$.
- For a specific p, given a $e_{j_i} \in ORD_p$ with $i \neq 1$, we refer to $e_{j_{i-1}}$ as $pred(e_{j_i})$.

We must also distinguish between the three types of stage, depending on their differences in their maintenance costs. This is a partition of the set $ETAPAS^*$. We can encounter "fixed maintenance" ($ETAPAS_f$), "binary maintenance" ($ETAPAS_b$) and "variable maintenance" ($ETAPAS_v$).

$$ETAPAS^* = ETAPAS_f \dot{\cup} ETAPAS_b \dot{\cup} ETAPAS_v$$

- The maintenance cost of the fixed stages does not depend on whether the stage is running or not. It also does not depend on the product or the production rate.
- Binary stages only incur a cost if the stage is active. The cost depends on the product but does not depend on the production rate.
- In the variable stages, maintenance cost is only dependent on the rate and product.

Another subset of the stages that we will have to define is "stages of initial cost" E_{ci} as all the stages with a cost when they are turned on.

Maintenance costs are given by the following matrixes:

- $Fijo = (F_e)$ with $e \in ETAPAS_f$, is the maintenance cost of the fixed stages.
- $Binario = (B_{p,e})$ con $p \in PRODS, e \in ETAPAS_b$, is the maintenance cost of the binary stages.
- $Variable = (V_{p,e})$ con $p \in PRODS, e \in ETAPAS_v$, is the maintenance cost of the variable stages when the rate is equal to 1.
- $CosteInicial = (CI_e)$ con $e \in ETAPAS_{ci}$ is the cost of turning on the stage e.

To sum up with our parameters, we define:

- The conversion factor from quantity of product to quantity of artichoke: $\Phi = (\Phi_p)$ with $p \in PRODS$, which is the amount of artichoke that is needed to produce p.

Variables: In this section, we specify and define the variables we will use along with their respective bounds.

- $c_{p,e,h,g}$ $\forall p \in PRODS, e \in ORD_p, h \in HORAS, g \in DIAS$: These represent the amount of product p waiting to be processed by the stage e at hour h of the day g. In this variable, we will use the auxiliary stage e_{fin} that we defined previously.

  The variable $c_{p,e_{fin},h,g}$ will be the product p already finished at hour h. Consequently, $c_{p,e_{fin},h_{max},g}$ is the product finished in one day.

  It is important to note that $c_{p,e_I(p),1,g}$ will be the quantity of artichokes that we allocate to produce p, which we decide at the beginning of the day. In this way, $0 \leq c_{p,e_I(p),1,g} \leq \frac{CantAlc}{\Phi_p}$.

  The rest of variables $c_{p,e,h,g}$ are tied to that of the first hour, so the bound is not strictly needed; nevertheless, as our rates are normalized, we can set:

$$0 \leq c_{p,e,h,g} \leq \frac{CantAlc}{\Phi_p} \quad \forall p \in PRODS, e \in ORD_p, h \in HORAS, g \in DIAS$$

- $r_{p,e,h,g}$ $\forall p \in PRODS, e \in ORD_p^*, h \in HORAS, g \in DIAS$: These are the amount of product p processed by the stage e at the hour h of day g. They are bounded by the parameters previously defined: $0 \leq r_{p,e,h,g} \leq R_{p,e}$.
- $w_{p,e,h,g}$ $\forall p \in PRODS, e \in ORD_p^*, h \in HORAS, g \in DIAS$: These represent whether if the stage e is producing p or ot at time h of day g. The variable is binary.
- $k_{e,h,g}$ $\forall e \in ETAPAS_{ci}, h \in HORAS, g \in DIAS$: It is binary and will be 1 if stage e has been turned on at hour h of day g. Be careful not to confuse this variable with the previous one. The variable does not indicate whether the stage is active at hour h, but whither it has been activated at that hour, that is if it was off at $h-1$ and now it is on.
- $s_{p,c}$ $\forall p \in PRODS, c \in CLS$: These represent the quantity of product p sold to costumer c at the end of the week. $D_{p,c} \leq s_{p,c} \leq M_{p,c}$.

Objective function: Our objective is to maximize profits, which we will calculate as the revenue from selling the products minus the maintenance costs of the stages and the cost of starting certain stages.

$$Maximize: \quad Ingresos - Mantenimiento - CosteInicio$$

Where:

- $Ingresos$ is the amount of money we earn from selling products.

$$Ingresos = \sum_{p \in PRODS, c \in CLS} s_{p,c} PR_{p,c}$$

  This is, the sum over the sales of a product times its selling price.

- $Mantenimiento$ is the amount of money spent due to the maintenance of the stages.

$$Mantenimiento = \sum_{e \in ETAPAS_f} F_e \cdot h_{max} \cdot g_{max} +$$

$$+ \sum_{g \in DIAS, p \in PRODS} \Bigg(\sum_{e \in E_b \cap ORD_p^*} \Big(\sum_{h \in HORAS} w_{p,e,h,g} \Big) B_{p,e} +$$

$$+ \sum_{e \in ETAPAS_v \cap ORD_p^*} \Big(\sum_{h \in HORAS} r_{p,e,h,g} \Big) V_{p,e} \Bigg)$$

  This term is divided into three parts, one for every type of stage in which we divided our original set of stages. Let us note that the first term, the cost of fixed stages, is constant, so we could not include it in the objective function. But we decided to include it to improve readability. The second and third terms are those associated to the binary and variable stages. The costs of the binary stages depend on the variable $w_{p,e,h}$ and the costs of the variable stages on the rate of production.

- $CosteInicio$ is the amount of money spent by starting a stage that was previously stopped.

$$CosteInicio = \sum_{e \in ETAPAS_{ci}, g \in DIAS} \Big(\sum_{h \in HORAS} k_{e,h,g} \Big) CI_e$$

  We calculate it as the start-up cost for each time the stage has to be turned on.

Constraints: In this section, we will specify the constraints we have established to properly model our problem.

Production constraints: We describe the constraints that affect production.

• The quantity we send for production cannot exceed the raw materials available, which we model as:

$$\sum_{p \in PRODS} c_{p,e_I(p),1,g} \Phi_p \leq CantAlc_g \quad \forall g \in DIAS$$

We need to convert the quantity of artichokes available into the quantity of product. Furthermore, we do this at the beginning of the day, as mentioned earlier.

• We must ensure that by the end of the day, nothing is left partially produced in the intermediate stages. Everything that has been introduced must be fully processed throughout the day. We model this with the following constraint:

$$c_{p,e_I(p),1,g} = c_{p,e_{fin},h_{max},g} \quad \forall p \in PRODS, g \in DIAS$$

The constraint achieves its purpose since these variables represent the initial and final quantity of product p, respectively.

• There cannot be partially completed products at the beginning of any day. That is, the quantity of products in a stage that is midway through the process in the first hour must be 0:

$$c_{p,e,1,g} = 0 \quad \forall p \in PRODS, e \in ORD_p \setminus \{e_I(p)\}, g \in DIAS$$

This constraint could be omitted by not creating these variables, but the notation for the subsequent ones would become somewhat more obtuse. For better clarity, we have decided to keep it this way.

• Production is neither created nor destroyed; it is only transported. The stages cannot generate products out of nothing, nor can they lose products (assuming no errors in the production line). It is modeled as:

$$c_{p,e,h,g} = c_{p,e,h-1,g} - r_{p,e,h-1,g} + r_{p,pred(e),h-1,g}$$
$$\forall p \in PRODS, e \in ORD_p^* \setminus \{e_I(p)\}, h \in HORAS, h > 1, g \in DIAS$$
$$c_{p,e_{fin},h,g} = c_{p,e_{fin},h-1,g} + r_{p,e_F(p),h-1,g} \quad \forall p \in PRODS, h \in HORAS, h > 1, g \in DIAS$$
$$c_{p,e_I(p),h,g} = c_{p,e_I(p),h-1,g} - r_{p,e_I(p),h-1,g} \quad \forall p \in PRODS, h \in HORAS, h > 1, g \in DIAS$$

That is, for each hour, the quantity of product in a stage must be equal to the quantity from the previous hour minus what the stage produced during that hour, plus what was produced by the previous stage and delivered to it.

Although it is divided into three constraints, they all mean the same thing; it is just that the initial and final stages are special cases where one of the rates is not considered.

• No stage can produce over its storage:

$$r_{p,e,h,g} \leq c_{p,e,h,g} \quad \forall p \in PRODS \ \forall e \in ORD_p^* \ \forall h \in HORAS, g \in DIAS$$

• If a stage is stopped, its rate must be zero. This constraint models the desired operation for the variables $w_{p,e,h,g}$.

$$(w_{p,e,h,g} = 0) \implies (r_{p,e,h,g} = 0) \quad \forall p \in PRODS, e \in ORD_p^*, h \in HORAS, g \in DIAS$$

This logical constraint will be linearized later.

• Only one product can be processed in each stage at the same time:

$$\sum_{p \in \Omega_e} w_{p,e,h,g} \leq 1 \quad \forall e \in ETAPAS^*, h \in HORAS, g \in DIAS$$

As the variables w are binary, this constraint guaranties that only one of them can be on at once.

Fixed Cost Constraint: In this case, the two constraints are not linear; we will linearize them later. Both of them model when a machine goes from being off to being on.

$$(\sum_{p \in \Omega_e} w_{p,e,1,g} = 1) \implies (k_{e,1,g} = 1) \quad \forall e \in ETAPAS_{ci}, g \in DIAS$$

• At the beginning of the day, at $h = 1$, the variable $k_{e,1,g}$ will simply be activated when some $w_{p,e,1,g}$ is also active. It is model as follows:

$$(\sum_{p \in \Omega_e} w_{p,e,1,g} = 1) \implies (k_{e,1,g} = 1) \quad \forall e \in ETAPAS_{ci}, g \in DIAS$$

• For any hour that is not the initial one, we need to check that some product is passing through the stage and that, in the previous hour, the stage was not occupied. This is:

$$(\sum_{p\in\Omega_e} w_{p,e,h,g} - \sum_{p\in\Omega_e} w_{p,e,h-1,g} = 1) \Longrightarrow (k_{e,h,g} = 1) \quad \forall h > 1, e \in ETAPAS_{ci}, g \in DIAS$$

We are only modeling one implication, since the variables $k_{e,h,g}$ affect negatively our objective function, so we can assume their default value to be 0.

Sales constraints:

• This constraint ensures that we cannot sell more or less than what we have produced.

$$\sum_{c\in CLS} s_{p,c} = \sum_{g\in DIAS} c_{p,e_{fin},h_{max},g} \quad \forall p \in PRODS$$

For each product, the sum of the quantities sold to each customer must be equal to the quantity produced.

Linearization: Some of the constraints that have arisen are not linear, so to make the problem easier to solve computationally, we will linearize them:

• For the constraint $(w_{p,e,h,g} = 0) \implies (r_{p,e,h,g} = 0)$ we set::

$$r_{p,e,h,g} \leq w_{p,e,h,g} N_{p,e} \quad \forall p \in PRODS, e \in ORD_p^*, h \in HORAS, g \in DIAS$$

Note that the rates $r_{p,e,h,g}$ are bounded by $R_{e,p}$, so it is convenient to take the big-M as: $N_{p,e} = R_{e,p}$.

• The other two constraints regarding the activation of the machines are linearized as follows:

$$\sum_{p\in\Omega_e} w_{p,e,1,g} \leq k_{e,1,g} \quad \forall e \in ETAPAS_{ci}, g \in DIAS$$

$$\sum_{p\in\Omega_e} w_{p,e,h,g} - \sum_{p\in\Omega_e} w_{p,e,h-1,g} \leq k_{e,h,g} \quad \forall h > 1, e \in ETAPAS_{ci}, g \in DIAS$$

Final formulation: By combining all the constraints, variables, and the objective function, our problem can be stated as:

$$
\begin{aligned}
\text{maximize} \quad & \sum_{p \in PRODS,\, c \in CLS} s_{p,c} PR_{p,c} - \sum_{e \in ETAPAS_f} F_e\, h_{max} g_{max} \\
& - \sum_{g \in DIAS, p \in PRODS} \Big(\sum_{e \in ETAPAS_b \cap ORD_p} \Big(\sum_{h \in HORAS} w_{p,e,h,g} \Big) B_{p,e} \\
& + \sum_{e \in ETAPAS_v \cap ORD_p} \Big(\sum_{h \in HORAS} r_{p,e,h,g} \Big) V_{p,e} \Big) \\
& - \sum_{e \in ETAPAS_{ci}, g \in DIAS} \Big(\sum_{h \in HORAS} k_{e,h,g} \Big) CI_e
\end{aligned}
$$

$$
\text{subject to} \quad \begin{cases}
\bullet \sum_{p \in PRODS} c_{p,e_I(p),1,g} \Phi_p \leq CantAlc_g \\
\forall g \in DIAS \\
\bullet c_{p,e_I(p),1,g} = c_{p,e_{fin},h_{max},g} \\
\forall p \in PRODS, g \in DIAS \\
\bullet c_{p,e,1,g} = 0 \\
\forall p \in PRODS, e \in ORD_p \setminus \{e_I(p)\}, g \in DIAS \\
\bullet c_{p,e,h,g} = c_{p,e,h-1,g} - r_{p,e,h-1,g} + r_{p,pred(e),h-1,g} \\
\forall p \in PRODS, e \in ORD_p^* \setminus \{e_I(p)\}, h \in HORAS, h > 1, g \in DIAS \\
\bullet c_{p,e_{fin},h,g} = c_{p,e_{fin},h-1,g} + r_{p,e_F(p),h-1,g} \\
\forall p \in PRODS, h \in HORAS, h > 1, g \in DIAS \\
\bullet c_{p,e_I(p),h,g} = c_{p,e_I(p),h-1,g} - r_{p,e_I(p),h-1,g} \\
\forall p \in PRODS, h \in HORAS, h > 1, g \in DIAS \\
\bullet r_{p,e,h,g} \leq c_{p,e,h,g} \\
\forall p \in PRODS\ \forall e \in ORD_p^*\ \forall h \in HORAS, g \in DIAS \\
\bullet \sum_{p \in \Omega_e} w_{p,e,h,g} \leq 1 \\
\forall e \in ETAPAS^*, h \in HORAS, g \in DIAS \\
\bullet r_{p,e,h,g} \leq w_{p,e,h,g} R_{p,e} \\
\forall p \in PRODS, e \in ORD_p^*, h \in HORAS, g \in DIAS \\
\bullet \sum_{p \in \Omega_e} w_{p,e,1,g} \leq k_{e,1,g} \\
\forall e \in ETAPAS_{ci}, g \in DIAS \\
\bullet \sum_{p \in \Omega_e} w_{p,e,h,g} - \sum_{p \in \Omega_e} w_{p,e,h-1,g} \leq k_{e,h,g} \\
\forall h > 1, e \in ETAPAS_{ci}, g \in DIAS \\
\bullet \sum_{c \in CLS} s_{p,c} = \sum_{g \in DIAS} c_{p,e_{fin},h_{max},g} \\
\forall p \in PRODS \\
\\
\bullet 0 \leq c_{p,e,h,g} \leq \frac{CantAlc_g}{\Phi_p} \\
\forall p \in PRODS, e \in ORD_p, h \in HORAS, g \in DIAS \\
\bullet 0 \leq r_{p,e,h,g} \leq R_{p,e} \\
\forall p \in PRODS, e \in ORD_p^*, h \in HORAS, g \in DIAS \\
\bullet D_{p,c} \leq s_{p,c} \leq M_{p,c} \\
\forall p \in PRODS, c \in CLS \\
\bullet w_{p,e,h,g} \in \{0,1\} \\
\forall p \in PRODS, e \in ORD_p^*, h \in HORAS, g \in DIAS \\
\bullet k_{e,h,g} \in \{0,1\} \\
\forall e \in ETAPAS_{ci}, h \in HORAS, g \in DIAS \\
\bullet c_{p,e,h,g}, r_{p,e,h,g}, s_{p,c} \in \mathbb{Z}^+
\end{cases}
$$

4 The Company's Specific Problem

In this section, we will present the company's data in order to later analyze the results provided by our model.

The factory operates 16 hours per day, 7 days a week, and the production line has the following stages, in the given order, with the following characteristics.

Stage	**Type**	**Inicial cost (CI)**
Machine peeling	Variable	No
Hand peeling	Fixed	No
Cutting	Binary	No
Confiting	Binary	Yes (82)
Placing in trays (Trays)	Fixed	No
Freezing	Variable	Yes (53)
Adding oil	Binary	No
Thermosealing	Binary	Yes (12)
Placing in boxes (Boxes)	Fixed	No

The company works with several artichoke sizes, ranging from 50 mm to 85 mm in steps of 5 mm. We will focus only on the 60 mm size, because it is the one with which they produce the greatest variety of products.

For the 60 mm size, we have the following 10 products:

- Trays with 15 pieces with and without oil ($B15_con$, $B15_sin$).
- Trays with 24 pieces with and without oil ($B24_con$, $B24_sin$).
- Trays with 6 pieces with and without oil ($B6_con$, $B6_sin$).
- Trays with 30 pieces with and without oil ($B30_con$, $B30_sin$).
- Raw pieces with and without oil (T_con, T_sin).

With this categorization, we can specify which stages each product will go through. The "Trays" skip the cutting stage, while the "Pieces" skip the confiting stage. The "without oil" products omit the adding oil stage.

Below, we present the characteristics of the different stages with respect to each product.

Fixed costs (Maintenance):

Stage	**Fixed costs**
Hand peeling	32
Placing in trays	16
Placing in boxes	20

Binary costs (Maintenance per product):

Stage	B15_con	B15_sin	B24_con	B24_sin	B6_con	B6_sin	B30_con	B30_sin	T_con	T_sin
Cutting	0	0	0	0	0	0	0	0	4	4
Confiting	10	10	10	10	10	10	10	10	0	0
Adding oil	5	0	5	0	5	0	5	0	5	0
Thermosealing	4	4	4	4	4	4	4	4	5	5

Variable costs (Per processed unit):

Stage	B15_con	B15_sin	B24_con	B24_sin	B6_con	B6_sin	B30_con	B30_sin	T_con	T_sin
Machine peeling	0.4	0.4	0.5	0.5	0.4	0.4	0.6	0.6	0.5	0.5
Freezing	1.0	1.0	1.1	1.1	0.9	0.9	1.2	1.2	1.0	1.0

The maximum real rates of the stages (units/hour) are:

Stage	B15_con	B15_sin	B24_con	B24_sin	B6_con	B6_sin	B30_con	B30_sin	T_con	T_sin
Machine peeling	5261	5261	5261	5261	5261	5261	5261	5261	5261	5261
Hand peeling	3046	3046	3046	3046	3046	3046	3046	3046	3046	3046
Cutting	0	0	0	0	0	0	0	0	3600	3600
Confiting	5537	5537	5537	5537	5537	5537	5537	5537	0	0
Trays	4984	4984	4153	4153	3322	3322	2769	2769	4984	4984
Freezing	6645	6645	6091	6091	5039	5039	4707	4707	7752	7752
Adding oil	11075	0	11075	0	11075	0	11075	0	11075	0
Thermosealing	5537	5537	5537	5537	5537	5537	5537	5537	5537	5537
Boxes	9967	9967	8859	8859	7752	7752	6645	6645	9414	9414

Necessary subproduct per stage:

Stage	B15_con	B15_sin	B24_con	B24_sin	B6_con	B6_sin	B30_con	B30_sin	T_con	T_sin
Trays	5	5	6	6	6	6	6	6	4	4
Boxes	3	3	4	4	1	1	5	5	5	5

Minimum demand per client:

Client	B15_con	B15_sin	B24_con	B24_sin	B6_con	B6_sin	B30_con	B30_sin	T_con	T_sin
C1	70	70	70	75	75	70	75	75	70	70
C2	80	40	80	75	85	80	45	75	35	30
C3	85	80	80	80	35	80	80	30	70	70
C4	15	18	18	20	20	30	20	25	25	25
C5	70	70	40	75	75	70	75	40	45	40
C6	10	0	10	0	15	0	15	0	7	0
C7	0	0	0	2	0	5	0	5	0	0
C8	0	1	0	3	0	0	5	0	0	5
C9	8	0	5	0	5	5	0	5	0	5
C10	0	2	0	0	0	3	0	0	0	0
C11	0	5	5	0	0	0	5	0	0	0
C12	0	0	5	0	5	0	0	5	0	0
C13	0	0	0	5	0	5	0	5	0	0
C14	7	10	0	0	15	0	0	15	10	0
C15	5	0	5	0	0	5	5	0	0	0

Maximum demand per client:

Client	B15_con	B15_sin	B24_con	B24_sin	B6_con	B6_sin	B30_con	B30_sin	T_con	T_sin
C1	109	104	101	107	113	104	107	109	101	101
C2	115	58	118	113	121	115	63	109	46	40
C3	124	118	121	115	52	118	115	46	98	98
C4	46	24	26	28	32	46	29	35	40	35
C5	107	101	58	104	109	107	108	58	63	58
C6	29	0	29	0	29	0	29	0	23	0
C7	0	0	0	15	0	15	0	15	0	0
C8	0	15	0	15	0	0	15	0	0	15
C9	20	0	15	0	15	20	0	20	0	15
C10	0	10	0	0	0	15	0	0	0	0
C11	0	15	20	0	0	0	20	0	0	0
C12	0	0	15	0	15	0	0	15	0	0
C13	20	0	0	15	0	20	0	15	0	0
C14	0	25	0	15	30	0	0	30	20	0
C15	15	0	15	0	0	15	15	0	0	0

Profit (Price) per unit:

Client	B6_con	B6_sin	B15_con	B15_sin	B24_con	B24_sin	B30_con	B30_sin	T_con	T_sin
C1	12	11	27	26	40	39	45	44	10	9
C2	11	11	26	0	39	38	0	44	0	0
C3	0	11	28	27	41	39	44	0	9	8
C4	11	0	25	24	38	37	43	43	0	0
C5	11	11	26	25	0	38	44	0	0	0
C6	10	0	24	23	37	0	42	42	8	7
C7	12	11	27	0	40	39	0	44	0	0
C8	11	0	25	25	38	38	43	0	9	8
C9	11	12	26	0	39	0	44	44	10	9
C10	0	10	24	23	0	36	43	0	0	0
C11	0	0	0	24	41	40	46	45	0	0
C12	11	10	0	24	38	0	0	43	0	0
C13	0	11	27	26	0	39	0	44	0	0
C14	12	0	0	27	40	0	0	46	10	9
C15	0	11	25	0	39	38	44	0	0	0

Amount of available raw material:

Day	**Amount of artichokes**
Monday	12,300
Tuesday	13,100
Wednesday	11,200
Thursday	9,800
Friday	10,500
Saturday	8,000
Sunday	7,000

5 AMPL Implementation

In this section we will show the source code of the weekly model for our implementation in AMPL.

5.1 MOD:

```
param H integer > 0; #Numero de Horas

set HORAS ordered := 1..H;
set PRODUCTOS;
set CLIENTES;
set DIAS;
set ETAPAS ordered;
set ETAPASCOMPLETO ordered:= ETAPAS union {card(ETAPAS) + 1};

param CantAlcach {DIAS} integer > 0; #Numero de cantidad de alcachofas
param RatioRealMax {PRODUCTOS,ETAPAS} >= 0;
#Ratio maximo de cada etapa por cada producto
param DemandaMin {PRODUCTOS,CLIENTES} >= 0;
#Demanda minima que solicita cada cliente de el producto
param DemandaMax {PRODUCTOS,CLIENTES} >=0;
#Demanda maxima que solicita cada cliente de el producto
param Precio {PRODUCTOS,CLIENTES} >= 0;
#Precio al que me compra el cliente cada producto

param TipoEtapas {ETAPAS}; #Parametro que dice el tipo de las etapas

set EtapasFijas within ETAPAS := {e in ETAPAS: TipoEtapas[e] = 1};
set EtapasBinarias within ETAPAS := {e in ETAPAS: TipoEtapas[e] = 2};
set EtapasVariables within ETAPAS := {e in ETAPAS: TipoEtapas[e] = 3};
# Las estapas que son de mantenimiento Fijo ,Binario y Variable

param MantFijas {EtapasFijas};
param MantBinarias {PRODUCTOS,EtapasBinarias};
param MantVariables {PRODUCTOS,EtapasVariables};
#La cantidad de dinero que hay que gastar para el mantenimiento de las estapas

param CosteInicial {ETAPAS};

set EtapasCoste within ETAPAS := {e in ETAPAS: CosteInicial[e] > 0};
#La cantidad de dinero que hay que gastar para activar cada etapa

param Receta {PRODUCTOS,ETAPAS};
set Orden {p in PRODUCTOS} within ETAPAS  ordered:= {e in ETAPAS: Receta[p,e]
    = 1};
set OrdenCompleto {p in PRODUCTOS}  ordered:= Orden[p] union {last(
    ETAPASCOMPLETO)};
#El orden que sigue cada producto de las etapas y una matriz auxiliar binaria

set OrdenInvs {e in ETAPAS} within PRODUCTOS := {p in PRODUCTOS: e in Orden[p
    ]};
#Orden inverso

param SubProdNecesario{PRODUCTOS,ETAPAS} >=0; #Para calcular los factores de
    conversion para la cantidad y los ratios a partir de la cantidad de
    subproducto
param FactorConversion_ratio {p in PRODUCTOS,e in Orden[p]} := prod{f in Orden
    [p]: ord(e) <= ord(f)} SubProdNecesario[p,f];
```

```
param FactorConversion {p in PRODUCTOS} := FactorConversion_ratio[p,first(Orden[p])]; #Factor de conversion de productos a alcachofas


param RatioMax {p in PRODUCTOS,e in Orden[p]} := RatioRealMax[p,e] / FactorConversion_ratio[p,e];  #Calculando Ratio Maximo normalizado a partir del real

/********************/

var Cantidad {p in PRODUCTOS, OrdenCompleto[p], HORAS, g in DIAS} integer >= 0, <= CantAlcach[g]/FactorConversion[p];
#Cantidad de producto que esta esperando en cierta etapa a una hora dada un dia determinado

var Ratio {p in PRODUCTOS,e in Orden[p],HORAS, DIAS} integer >=0, <=RatioMax[p,e];
#Cantidad de producto que procesa una etapa en una hora dada un dia determinado

var Ocupado {p in PRODUCTOS,Orden[p],HORAS, DIAS} binary;
#1 si el producto dado esta en la etapa dada en la hora dada el dia indicado

var Encendido {EtapasCoste,HORAS, DIAS} binary;
#1 si la etapa dada esta encendida en la hora dada el dia indicado

var Vender {p in PRODUCTOS,c in CLIENTES} integer >= DemandaMin[p,c], <= DemandaMax[p,c];
#Cantidad de producto vendido a un cliente

/*******************/

maximize GanarDinero:
    (sum{p in PRODUCTOS, c in CLIENTES} (Vender[p,c]*Precio[p,c]))
    - (sum{e in EtapasFijas, g in DIAS} (MantFijas[e]*card(HORAS)))
    - (sum {p in PRODUCTOS, g in DIAS} ((sum {e in (EtapasBinarias inter Orden[p])}(sum {h in HORAS} Ocupado[p,e,h,g]) * MantBinarias[p,e])
    + (sum {e in (EtapasVariables inter Orden[p])}(sum {h in HORAS} Ratio[p,e,h,g]) * MantVariables[p,e])))
    - (sum{e in EtapasCoste, h in HORAS, g in DIAS} Encendido[e,h,g]*CosteInicial[e]);
# Esta es nuestra funcion objetivo

/*******************/

subject to Cota_materia {g in DIAS}:
    sum{p in PRODUCTOS} Cantidad[p,first(Orden[p]),1,g]*FactorConversion[p] <= CantAlcach[g];
# Cantidad maxima de producto que se puede usar al dia

subject to Final_es_inicial {p in PRODUCTOS, g in DIAS}:
    Cantidad[p,first(Orden[p]),1,g] = Cantidad[p,last(ETAPASCOMPLETO),H,g];
# La cantidad de alcachofa producir ha de ser la misma cantidad para productos

subject to Inicializacion {p in PRODUCTOS, e in (OrdenCompleto[p] diff {first(Orden[p])}), g in DIAS}:
    Cantidad[p,e,1,g] = 0;
# Al comienzo del dia las maquinas empiezan vacias

subject to Produccion_crear {p in PRODUCTOS, e in (Orden[p] diff {first(Orden[p])}), h in (HORAS diff {1}), g in DIAS}:
```

```
    Cantidad[p,e,h,g] = Cantidad[p,e,h-1,g] - Ratio[p,e,h-1,g] + Ratio[p,prev(e,Orden[p]),h-1,g];
# Restriccion que liga la cadena de produccion y hace que no se cree producto en etapas intermedias

subject to Produccion_crear_final {p in PRODUCTOS, h in (HORAS diff {1}), g in DIAS}:
    Cantidad[p,last(ETAPASCOMPLETO),h,g] = Cantidad[p,last(ETAPASCOMPLETO),h-1,g] + Ratio[p,last(Orden[p]),h-1,g];
# Restriccion que acumula al final de la cadena de produccion todos los productos ya terminados cada dia

subject to Produccion_crear_inicial {p in PRODUCTOS, h in (HORAS diff {1}),g in DIAS}:
    Cantidad[p,first(Orden[p]),h,g] = Cantidad[p,first(Orden[p]),h-1,g] - Ratio[p,first(Orden[p]),h-1, g];
# Restriccion que va tomando las alcachofas al principio de la cadena para empezar a procesarlas por horas cada

subject to Producir_Posible {p in PRODUCTOS, e in Orden[p], h in HORAS, g in DIAS}:
    Ratio[p,e,h,g] <= Cantidad[p,e,h,g];
# El ratio de produccion de una hora no puede ser mas grande que
# la cantidad de productos que se tienen esperando en esa etapa

subject to Unicidad_producto {e in ETAPAS, h in HORAS, g in DIAS}:
    sum{p in OrdenInvs[e]} Ocupado[p,e,h,g] <= 1;
# Cada etapa solo puede estar trabajando un producto a la vez

subject to Cota_ratio {p in PRODUCTOS, e in Orden[p], h in HORAS, g in DIAS}:
    Ratio[p,e,h,g] <= Ocupado[p,e,h,g]*RatioMax[p,e];
# Si la maquina esta encendida, el ratio no puede ser mayor que su maximo asociado
# a cada etapa en cada hora y para cada producto en cada dia

subject to Encender_si_produciendo_ini {e in EtapasCoste, g in DIAS}:
    sum{p in OrdenInvs[e]} Ocupado[p,e,1,g] <= Encendido[e,1,g];
# Restriccion que cuenta si la etapa inicial de cada producto ha pasado de inactiva a activa

subject to Encender_si_produciendo {e in EtapasCoste, h in (HORAS diff {1}), g in DIAS}:
    sum{p in OrdenInvs[e]} (Ocupado[p,e,h,g] - Ocupado[p,e,h-1,g]) <= Encendido[e,h,g];
# Restriccion que cuenta si una etapa distinta de la primera acaba de empezar a producir

subject to Rest_venta {p in PRODUCTOS}:
    (sum{c in CLIENTES} Vender[p,c]) <= (sum{g in DIAS} Cantidad[p,last(ETAPASCOMPLETO),H,g]);
# Restriccion que vende todo el producto que se ha creado
```

5.2 DAT:

```
# --- PARAMETROS GLOBALES ---
param H := 16;

# --- CONJUNTOS ---
set PRODUCTOS :=
    B15_con B15_sin B24_con B24_sin B6_con B6_sin B30_con B30_sin T_con T_sin;

set CLIENTES :=
    C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12 C13 C14 C15;

set DIAS :=
    Lunes Martes Miercoles Jueves Viernes Sabado Domingo;

set ETAPAS :=
    PeladoMaquina PeladoMano Cortado Confitadora Bandejas Congelacion Aceite
        Termosellado Cajas;

# --- CONFIGURACION DE ETAPAS ---
param TipoEtapas :=
PeladoMaquina 3
PeladoMano    1
Cortado       2
Confitadora   2
Bandejas      1
Congelacion   3
Aceite        2
Termosellado  2
Cajas         1;

# --- COSTES DE ACTIVACION ---
param CosteInicial :=
PeladoMaquina 0
PeladoMano 0
Cortado 0
Confitadora 82
Bandejas 0
Congelacion 53
Aceite 0
Termosellado 12
Cajas 0;

# --- MANTENIMIENTOS ---
param MantFijas :=
PeladoMano 32
Bandejas  16
Cajas 20;

param MantBinarias : Cortado Confitadora Aceite Termosellado :=
B15_con   0   10   5   4
B15_sin   0   10   0   4
B24_con   0   10   5   4
B24_sin   0   10   0   4
B6_con    0   10   5   4
B6_sin    0   10   0   4
B30_con   0   10   5   4
B30_sin   0   10   0   4
T_con     4    0   5   5
T_sin     4    0   0   5;

```

```
param MantVariables : PeladoMaquina Congelacion :=
B15_con   0.4   1.0
B15_sin   0.4   1.0
B24_con   0.5   1.1
B24_sin   0.5   1.1
B6_con    0.4   0.9
B6_sin    0.4   0.9
B30_con   0.6   1.2
B30_sin   0.6   1.2
T_con     0.5   1.0
T_sin     0.5   1.0;

# --- MATERIA PRIMA ---
param CantAlcach :=
Lunes 12300
Martes 13100
Miercoles 11200
Jueves 9800
Viernes 10500
Sabado 8000
Domingo 7000;

# --- RECETA  ---
param Receta:
PeladoMaquina PeladoMano Cortado Confitadora Bandejas Congelacion Aceite
    Termosellado Cajas :=
B15_con   1 1 0 1 1 1 1 1 1
B15_sin   1 1 0 1 1 1 0 1 1
B24_con   1 1 0 1 1 1 1 1 1
B24_sin   1 1 0 1 1 1 0 1 1
B6_con    1 1 0 1 1 1 1 1 1
B6_sin    1 1 0 1 1 1 0 1 1
B30_con   1 1 0 1 1 1 1 1 1
B30_sin   1 1 0 1 1 1 0 1 1
T_con     1 1 1 0 1 1 1 1 1
T_sin     1 1 1 0 1 1 0 1 1;

# --- RATIOS MAXIMOS ---

param RatioRealMax:
        PeladoMaquina PeladoMano Cortado Confitadora Bandejas Congelacion
            Aceite Termosellado Cajas :=

B15_con   5261 3046 0   5537 4984 6645 11075 5537 9967
B15_sin   5261 3046 0   5537 4984 6645 0     5537 9967
B24_con   5261 3046 0   5537 4153 6091 11075 5537 8859
B24_sin   5261 3046 0   5537 4153 6091 0     5537 8859
B6_con    5261 3046 0   5537 3322 5039 11075 5537 7752
B6_sin    5261 3046 0   5537 3322 5039 0     5537 7752
B30_con   5261 3046 0   5537 2769 4707 11075 5537 6645
B30_sin   5261 3046 0   5537 2769 4707 0     5537 6645
T_con     5261 3046 3600 0   4984 7752 11075 5537 9414
T_sin     5261 3046 3600 0   4984 7752 0     5537 9414;

# --- SUBPRODUCTO NECESARIO ACTUALIZADO ---
param SubProdNecesario:
PeladoMaquina PeladoMano Cortado Confitadora Bandejas Congelacion Aceite
    Termosellado Cajas :=
B15_con   1 1 0 1 5 1 1 1 3
B15_sin   1 1 0 1 5 1 0 1 3
B24_con   1 1 0 1 6 1 1 1 4
B24_sin   1 1 0 1 6 1 0 1 4
```

```
B6_con     1 1 0 1 6 1 1 1 1
B6_sin     1 1 0 1 6 1 0 1 1
B30_con    1 1 0 1 6 1 1 1 5
B30_sin    1 1 0 1 6 1 0 1 5
T_con      1 1 1 0 4 1 1 1 5
T_sin      1 1 1 0 4 1 0 1 5;

# --- MERCADO ---
param Precio:
        C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12 C13 C14 C15 :=
B6_con  12 11 0 11 11 10 12 11 11 0 0 11 0 12 0
B6_sin  11 11 11 0 11 0 11 0 12 10 0 10 11 0 11
B15_con 27 26 28 25 26 24 27 25 26 24 0 0 27 0 25
B15_sin 26 0 27 24 25 23 0 25 0 23 24 24 26 27 0
B24_con 40 39 41 38 0 37 40 38 39 0 41 38 0 40 39
B24_sin 39 38 39 37 38 0 39 38 0 36 40 0 39 0 38
B30_con 45 0 44 43 44 42 0 43 44 43 46 0 0 0 44
B30_sin 44 44 0 43 0 42 44 0 44 0 45 43 44 46 0
T_con   10 0 9 0 0 8 0 9 10 0 0 0 0 10 0
T_sin   9 0 8 0 0 7 0 8 9 0 0 0 0 9 0 ;


param DemandaMin:
           C1   C2   C3   C4   C5   C6   C7   C8   C9   C10 C11 C12 C13 C14 C15 :=
B15_con    70   80   85   15   70   10    0    0    8    0   0   0   7   0   5
B15_sin    70   40   80   18   70    0    0    1    0    2   5   0   0  10   0
B24_con    70   80   80   18   40   10    0    0    5    0   5   5   0   0   5
B24_sin    75   75   80   20   75    0    2    3    0    0   0   0   5   0   0
B6_con     75   85   35   20   75   15    0    0    5    0   0   5   0  15   0
B6_sin     70   80   80   30   70    0    5    0    5    3   0   0   5   0   5
B30_con    75   45   80   20   75   15    0    5    0    0   5   0   0   0   5
B30_sin    75   75   30   25   40    0    5    0    5    0   0   5   5  15   0
T_con      70   35   70   25   45    7    0    0    0    0   0   0   0  10   0
T_sin      70   30   70   25   40    0    0    5    5    0   0   0   0   0   0 ;

param DemandaMax:
           C1   C2   C3   C4   C5   C6   C7   C8   C9   C10 C11 C12 C13 C14 C15 :=

B15_con   109  115  124  46  107  29    0    0   20    0   0   0  20   0  15
B15_sin   104   58  118  24  101   0    0   15    0   10  15   0   0  25   0
B24_con   101  118  121  26   58  29    0    0   15    0  20  15   0   0  15
B24_sin   107  113  115  28  104   0   15   15    0    0   0   0  15   0   0
B6_con    113  121   52  32  109  29    0    0   15    0   0  15   0  30   0
B6_sin    104  115  118  46  107   0   15    0   20   15   0   0  20   0  15
B30_con   107   63  115  29  108  29    0   15    0    0  20   0   0   0  15
B30_sin   109  109   46  35   58   0   15    0   20    0   0  15  15  30   0
T_con     101   46   98  40   63  23    0    0    0    0   0   0   0  20   0
T_sin     101   40   98  35   58   0    0   15   15    0   0   0   0   0   0 ;
```

5.3 RUN:

```
reset;
model "ProduccionL3P.mod";
data "ProduccionL3P.dat";

option solver gurobi;
option gurobi_options 'timelim=300 outlev=1';
/*
Como nuestro problema es computacionalmente costoso, vamos a quedarnos
```

```
con la mejor soluci n que nos puede dar la maquina en 5 minutos
*/
solve;
display _solve_elapsed_time;

print("Crearemos las siguientes cantidades de producto para cada tipo distinto de ellos:");
print("Crearemos las siguientes cantidades de producto para cada tipo distinto de ellos:") >> SalidaDatos.txt;
for {p in PRODUCTOS} {
    printf("Producto ");
    printf("Producto ") >> SalidaDatos.txt;
    printf(p);
    printf(p) >> SalidaDatos.txt;
    printf(" || ");
    printf(" || ") >> SalidaDatos.txt;
    printf("\n");
    printf("\n") >> SalidaDatos.txt;

    for {g in DIAS} {
    printf(" dia ");
    printf(" dia ") >> SalidaDatos.txt;
    printf(g);
    printf(g) >> SalidaDatos.txt;
    printf(": ");
    printf(": ") >> SalidaDatos.txt;
    printf(Cantidad[p,last(OrdenCompleto[p]),H,g]);
    printf(Cantidad[p,last(OrdenCompleto[p]),H,g]) >> SalidaDatos.txt;
    printf("\n");
    printf("\n") >> SalidaDatos.txt;
    }

    printf("\n");
    printf("\n") >> SalidaDatos.txt;

}

printf("\n");
print("-----------------------------------------");
printf("\n");

printf("\n") >> SalidaDatos.txt;
print("-----------------------------------------") >> SalidaDatos.txt;
printf("\n") >> SalidaDatos.txt;

print("Le venderemos a los clientes las siguientes cantidades:");
print("Le venderemos a los clientes las siguientes cantidades:") >> SalidaDatos.txt;
for {p in PRODUCTOS} {
    printf("Producto ");
    printf("Producto ") >> SalidaDatos.txt;
    printf(p);
    printf(p) >> SalidaDatos.txt;
    printf(" || ");
    printf(" || ") >> SalidaDatos.txt;
    printf("\n");
    printf("\n") >> SalidaDatos.txt;

    for {c in CLIENTES} {
        printf(" cliente ");
        printf(" cliente ") >> SalidaDatos.txt;
        printf(c);
```

```
        printf(c) >> SalidaDatos.txt;
        printf(": ");
        printf(": ") >> SalidaDatos.txt;
        printf(Vender[p,c]);
        printf(Vender[p,c]) >> SalidaDatos.txt;
        printf("\n");
        printf("\n") >> SalidaDatos.txt;
    }

    printf("\n");
    printf("\n") >> SalidaDatos.txt;

}

printf("\n");
print("-_-_-_
```

```
            print("--------------------------------------------------") >>
                SalidaDatos.txt;
        }
    }

    print("");
    print("") >> SalidaDatos.txt;

}
```

6 Conclusion

After running the code, the program provided us with the following solution:

	Value
Total benefits	80,285.20 €

Product	**Mon**	**Tue**	**Wed**	**Thu**	**Fri**	**Sat**	**Sun**	**TOTAL**
B15_con	284	0	281	0	0	0	0	**565**
B15_sin	0	110	0	0	0	322	0	**432**
B24_con	0	0	126	337	0	0	0	**463**
B24_sin	0	0	165	0	0	132	175	**472**
B6_con	0	488	0	0	0	0	0	**488**
B6_sin	0	547	0	0	0	0	0	**547**
B30_con	268	0	0	57	0	0	0	**325**
B30_sin	0	0	0	0	280	0	0	**280**
T_con	0	262	0	0	0	0	0	**262**
T_sin	0	0	0	0	105	0	140	**245**

Table 1: Daily production quantities and weekly totals

Prod.	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**	**C7**	**C8**	**C9**	**C10**	**C11**	**C12**	**C13**	**C14**	**C15**	**TOTAL**
B15_c	109	115	124	46	107	10	0	0	20	0	0	0	20	0	14	**565**
B15_s	104	40	118	18	101	0	0	15	0	2	9	0	0	25	0	**432**
B24_c	101	118	121	18	40	10	0	0	15	0	20	5	0	0	15	**463**
B24_s	107	110	115	20	75	0	15	15	0	0	0	0	15	0	0	**472**
B6_c	113	121	35	32	109	18	0	0	15	0	0	15	0	30	0	**488**
B6_s	104	115	118	30	107	0	15	0	20	3	0	0	20	0	15	**547**
B30_c	75	45	80	20	75	15	0	5	0	0	5	0	0	0	5	**325**
B30_s	75	75	30	25	40	0	5	0	5	0	0	5	5	15	0	**280**
T_c	70	35	70	25	45	7	0	0	0	0	0	0	0	10	0	**262**
T_s	70	30	70	25	40	0	0	5	5	0	0	0	0	0	0	**245**

Table 2: Sales breakdown by customer and validation of total sales.

The daily production is organized as follows:

Table 3: Hourly production planning: Product B15_con

Day	**Hour**	**Stage**	**Normalized Rate**	**Real Rate (u/h)**
Monday	1	Machine peeling	284	4260
Monday	2	Hand Peeling	81	1215
Monday	5	Hand Peeling	203	3045
Monday	6	Confiting	284	4260
Monday	7	Trays	284	4260
Monday	8	Freezing	284	852
Monday	10	Adding oil	284	852
Monday	11	Thermosealing	284	852
Monday	15	Boxes	284	852
Wednesday	2	Machine peeling	281	4215
Wednesday	3	Hand Peeling	78	1170
Wednesday	4	Hand Peeling	203	3045
Wednesday	5	Confiting	281	4215
Wednesday	9	Trays	281	4215
Wednesday	10	Freezing	281	843
Wednesday	11	Adding oil	281	843
Wednesday	13	Thermosealing	281	843
Wednesday	15	Boxes	281	843

Table 4: Hourly production planning: Product B15_sin

Day	**Hour**	**Stage**	**Normalized Rate**	**Real Rate (u/h)**
Tuesday	4	Machine peeling	110	1650
Tuesday	5	Hand Peeling	110	1650
Tuesday	7	Confiting	110	1650
Tuesday	8	Trays	110	1650
Tuesday	9	Freezing	110	330
Tuesday	10	Thermosealing	110	330
Tuesday	13	Boxes	110	330
Saturday	1	Machine peeling	322	4830
Saturday	2	Hand Peeling	203	3045
Saturday	3	Hand Peeling	119	1785
Saturday	7	Confiting	322	4830
Saturday	8	Trays	223	3345
Saturday	10	Trays	99	1485
Saturday	11	Freezing	322	966
Saturday	12	Thermosealing	322	966
Saturday	15	Boxes	322	966

Table 5: Hourly production planning: Product B24_con

Day	**Hour**	**Stage**	**Normalized Rate**	**Real Rate (u/h)**
Wednesday	1	Machine peeling	126	3024
Wednesday	2	Hand Peeling	126	3024
Wednesday	6	Confiting	126	3024
Wednesday	7	Trays	126	3024
Wednesday	9	Freezing	126	504
Wednesday	10	Adding oil	126	504
Wednesday	11	Thermosealing	126	504
Wednesday	12	Boxes	126	504
Thursday	1	Machine peeling	118	2832
Thursday	2	Hand Peeling	104	2496
Thursday	3	Machine peeling	219	5256
Thursday	4	Hand Peeling	126	3024
Thursday	5	Hand Peeling	107	2568
Thursday	5	Confiting	230	5520
Thursday	6	Confiting	107	2568
Thursday	7	Trays	173	4152
Thursday	8	Freezing	173	692
Thursday	9	Trays	164	3936
Thursday	10	Freezing	164	656
Thursday	11	Adding oil	337	1348
Thursday	12	Thermosealing	337	1348
Thursday	15	Boxes	337	1348

Table 6: Hourly production planning: Product B24_sin

Day	Hour	Stage	Normalized Rate	Real Rate (u/h)
Wednesday	4	Machine peeling	165	3960
Wednesday	5	Hand Peeling	39	936
Wednesday	6	Hand Peeling	126	3024
Wednesday	7	Confiting	165	3960
Wednesday	10	Trays	165	3960
Wednesday	11	Freezing	165	660
Wednesday	12	Thermosealing	165	660
Wednesday	13	Boxes	165	660
Saturday	2	Machine peeling	6	144
Saturday	4	Machine peeling	126	3024
Saturday	4	Hand Peeling	6	144
Saturday	5	Hand Peeling	126	3024
Saturday	6	Confiting	132	3168
Saturday	7	Trays	132	3168
Saturday	12	Freezing	132	528
Saturday	13	Thermosealing	132	528
Saturday	14	Boxes	132	528
Sunday	2	Machine peeling	49	1176
Sunday	3	Machine peeling	126	3024
Sunday	3	Hand Peeling	49	1176
Sunday	4	Hand Peeling	126	3024
Sunday	5	Confiting	175	4200
Sunday	6	Trays	173	4152
Sunday	7	Trays	2	48
Sunday	8	Freezing	175	700
Sunday	9	Thermosealing	175	700
Sunday	12	Boxes	175	700

Table 7: Hourly production planning: Product B6_con

Day	Hour	Stage	Normalized Rate	Real Rate (u/h)
Tuesday	1	Machine peeling	488	2928
Tuesday	6	Hand Peeling	488	2928
Tuesday	8	Confiting	488	2928
Tuesday	9	Trays	488	2928
Tuesday	10	Freezing	488	488
Tuesday	11	Adding oil	488	488
Tuesday	12	Thermosealing	488	488
Tuesday	14	Boxes	488	488

Table 8: Hourly production planning: Product B6_sin

Day	**Hour**	**Stage**	**Normalized Rate**	**Real Rate (u/h)**
Tuesday	6	Machine peeling	547	3282
Tuesday	7	Hand Peeling	507	3042
Tuesday	8	Hand Peeling	40	240
Tuesday	9	Confiting	547	3282
Tuesday	11	Trays	547	3282
Tuesday	12	Freezing	547	547
Tuesday	13	Thermosealing	547	547
Tuesday	15	Boxes	547	547

Table 9: Hourly production planning: Product B30_con

Day	**Hour**	**Stage**	**Normalized Rate**	**Real Rate (u/h)**
Monday	2	Machine peeling	101	3030
Monday	3	Machine peeling	167	5010
Monday	3	Hand Peeling	101	3030
Monday	4	Hand Peeling	101	3030
Monday	5	Confiting	92	2760
Monday	6	Hand Peeling	66	1980
Monday	6	Trays	92	2760
Monday	7	Confiting	176	5280
Monday	8	Trays	84	2520
Monday	9	Trays	92	2760
Monday	10	Freezing	268	1340
Monday	11	Adding oil	268	1340
Monday	12	Thermosealing	268	1340
Monday	13	Boxes	268	1340
Thursday	2	Machine peeling	57	1710
Thursday	3	Hand Peeling	57	1710
Thursday	7	Confiting	57	1710
Thursday	8	Trays	57	1710
Thursday	9	Freezing	57	285
Thursday	10	Adding oil	57	285
Thursday	11	Thermosealing	57	285
Thursday	14	Boxes	57	285

Table 10: Hourly production planning: Product B30_con

Day	Hour	Stage	Normalized Rate	Real Rate (u/h)
Monday	2	Machine peeling	101	3030
Monday	3	Machine peeling	167	5010
Monday	3	Hand Peeling	101	3030
Monday	4	Hand Peeling	101	3030
Monday	5	Confiting	92	2760
Monday	6	Hand Peeling	66	1980
Monday	6	Trays	92	2760
Monday	7	Confiting	176	5280
Monday	8	Trays	84	2520
Monday	9	Trays	92	2760
Monday	10	Freezing	268	1340
Monday	11	Adding oil	268	1340
Monday	12	Thermosealing	268	1340
Monday	13	Boxes	268	1340
Thursday	2	Machine peeling	57	1710
Thursday	3	Hand Peeling	57	1710
Thursday	7	Confiting	57	1710
Thursday	8	Trays	57	1710
Thursday	9	Freezing	57	285
Thursday	10	Adding oil	57	285
Thursday	11	Thermosealing	57	285
Thursday	14	Boxes	57	285

Table 11: Hourly production planning: Product T_con

Day	Hour	Stage	Normalized Rate	Real Rate (u/h)
Tuesday	2	Machine peeling	262	5240
Tuesday	3	Hand Peeling	110	2200
Tuesday	4	Hand Peeling	152	3040
Tuesday	5	Cutting	180	3600
Tuesday	6	Cutting	82	1640
Tuesday	6	Trays	13	260
Tuesday	7	Trays	249	4980
Tuesday	8	Freezing	262	1310
Tuesday	9	Adding oil	262	1310
Tuesday	11	Thermosealing	262	1310
Tuesday	12	Boxes	262	1310

Table 12: Hourly production planning: Product T_sin

Day	**Hour**	**Stage**	**Normalized Rate**	**Real Rate (u/h)**
Friday	3	Machine peeling	105	2100
Friday	4	Hand Peeling	105	2100
Friday	5	Cutting	105	2100
Friday	6	Trays	105	2100
Friday	12	Freezing	105	525
Friday	13	Thermosealing	105	525
Friday	14	Boxes	105	525
Sunday	1	Machine peeling	140	2800
Sunday	2	Hand Peeling	140	2800
Sunday	3	Cutting	140	2800
Sunday	4	Trays	140	2800
Sunday	5	Freezing	140	700
Sunday	8	Thermosealing	140	700
Sunday	13	Boxes	140	700